LESSON 15
SPECIAL DIVISION

SUMMARY

15.1 Division by 9
15.2 Division by 8 etc.
15.3 Division by 99, 98 etc.
15.4 Divisor Below a Base Number
15.5 Divisor Above a Base Number

15.1 DIVISION BY 9

As you have seen before, the number 9 is special and there is a very easy way to divide by 9.

1 Find **23 ÷ 9**.

9)2 3
2 r 5

The first figure of 23 is the answer: 2.
And adding the figures of 23 gives the remainder: 2 + 3 = 5.

So **23 ÷ 9 = 2 remainder 5**.

It is easy to see why this works because every 10 contains a 9 with 1 left over.
So 2 tens contains **2** nines with **2** left over.
The answer is the same as the remainder, 2.
And that is why we add 2 to 3 to get the remainder.

✐ **Practice A** Divide by 9:

a 9)51 **b** 9)34 **c** 9)17 **d** 9)44

e 9)60 **f** 9)71 **g** 9)26 **h** 9)46

---

a 5 r6 b 3 r7 c 1 r8 d 4 r8
e 6 r6 f 7 r8 g 2 r8 h 4 r10 = 5 r1

It can happen that there is another nine in the remainder, as occurred in the last question of the last exercise and as the next example shows.

2 Find **66 ÷ 9**.

9)6 6
6 r 12 = **7 r 3**

We get 6 remainder 12 and there is another nine in the remainder of 12.

So we add the one extra nine to the 6, which becomes 7.
And the remainder is reduced to 3 (take 9 from 12).

3 Find **58 ÷ 9.**

9)5 8
5 r 13 = **6 r 4**

Remember you are trying to find the number of nines in 66 and the first answer you get is 6 remainder 12. So there are 6 nines with 12 remaining. Since there is another nine in the 12 you therefore have 7 nines altogether and 3 remaining.

You can also get the final remainder, 3, by adding the digits in 12.

✐ **Practice B** Divide the following by 9:

a 9)57 **b** 9)77 **c** 9)58 **d** 9)49

e 9)64 **f** 9)88 **g** 9)96

---

a 6 r3 b 8 r5 c 6 r4 d 5 r4
e 7 r1 f 9 r7 g 10 r6

The unique property of number nine, that it is one unit below ten leads to many of the very easy Vedic methods, as in Sections 15.2, 15.3, 15.4 following. See also the methods of converting fractions to their recurring decimal form in Manual 2 (or References 1 and 3), as well as corresponding algebraic applications.

LONGER NUMBERS

This can be easily extended to longer numbers.

4 **2301 ÷ 9 = 255 remainder 6**.

The sum can be set out like this: 9) 2 3 0 1

The **2** at the beginning of 2301 is brought straight down into the answer:

9) 2 3 0 1
2

This 2 is then added to the 3 in 2301, and **5** is put down:

9) 2 3 0 1
2 5

This 5 is then added to the 0 in 2301, and **5** is put down:

9) 2 3 0 1
2 5 5

This 5 is then added to 1 to give the remainder, **6**:

9) 2 3 0 1
2 5 5 r6

The first figure of the number being divided is the first figure of the answer, and each figure in the answer is added to the next figure in the dividend to give the next figure of the answer.
The last number we write down is the remainder.

✐ **Practice C** Divide the following:

a 9)2 1 2 3 **b** 9)3 1 2 **c** 9)1 1 2 0 2

d 9)4 3 1 **e** 9)5 0 3 0 **f** 9)7 0 7

g 9)2 0 3 0 1 0 **h** 9)1 6 4 **i** 9)3 1 0 3 2

a	**235 r8**	**b**	**34 r6**	**c**	**1244 r6**
d	**47 r8**	**e**	**558 r8**	**f**	**78 r5**
g	**22556 r6**	**h**	**18 r2**	**i**	**3448 r0**

CARRIES

In the method of division by 9 which you have used it can happen that a 2-figure number appears in the answer.

5 Find **3172 ÷ 9**.

```
9) 3   1   7    2
   3   4  1 1 r 13
```

= **352 remainder 4**

Here you find you get an 11 and a 13:
the first 1 in the 11 must be carried over to the 4, giving 351,
and there is also another 1 in the remainder so we get **352** remainder **4**.

✐ **Practice D** Divide the following:

a 9)6 1 5 3 **b** 9)3 2 8 2 **c** 9)5 5 5

d 9)8 2 5 2 **e** 9)6 6 1 **f** 9)4 7 4 1

g 9)1 2 3 4 5 **h** 9)4 7 4 7 **i** 9)2 0 0 8 2

a	**683 rem 6**	**b**	**364 rem 6**	**c**	**61 rem 6**
d	**916 rem 8**	**e**	**73 rem 4**	**f**	**526 rem 7**
g	**1371 r6**	**h**	**527 r4**	**i**	**2231 r3**

A SHORT CUT

We can avoid the double figures that crop up in some of these sums.
Let us do Example 5 above again.

6 Find **3172 ÷ 9**.

We can avoid the build-up of large numbers like 11 and 13.
In the last example we may notice, before we put the 4 down, that the next step will give a 2-figure number and so we put 5 down instead:

```
9) 3   1   7   2
   3   5   2 r 4
```

Then add 5 to 7 to get 12, but as the 1 has already been carried over we only put the 2 down. Finally, 2+2 = 4.

7 Find **777 ÷ 9**.

9) 7 7 7
8 6 r 3

If we put 7 for the first figure we get 14 at the next step, so we put 8.
8+7 = 15 and the 1 has already been carried over.

Now, if we put the 5 down we see a 2-figure number coming in the next step, so we put 6 down.

6+7 = 13 and the 1 has been carried over, so just put down the 3.

✐ **Practice E** Divide the following by 9:

a 6153	**b** 3272	**c** 555	**d** 8252
e 661	**f** 4741	**g** 5747	**h** 2938
i 12345	**j** 75057	**k** 443322	**l** 1918161

a 683 rem 6	**b 363 rem 5**	**c 61 rem 6**	**d 916 rem 8**
e 73 rem 4	**f 526 rem 7**	**g 638 rem 5**	**h 326 rem 4**
i 1371 rem 6	**j 8339 rem 6**	**k 49258**	**l 213129**

15.2 DIVISION BY 8 ETC.

This easy way to divide by 9 can be extended for 8, 7 etc.

8 Suppose we want to divide **31** by **8**.

8) 3 1
3 r 7

We bring the first **3** down into the answer.
Then instead of adding this to the 1 as we do when dividing by 9,
we add double 3 to the 1 to get **7** for the remainder.

We **double** the 3 because 8 is **2** below 10.

✐ **Practice F** Try some of these:

a 8) 2 2 **b** 8) 1 5 **c** 8) 2 5 **d** 8) 5 1

---

a 2 r6 b 1 r7 c 3 r1 d 6 r3

9 Similarly for **211** divided by **8**:

8) 2 1 1
2 5 r 11 = 26 r3

We bring down the first 2,
add double this to the 1 in the next column and put down 5,
then add double the 5 to the 1 in the last column and put down 11 as the remainder.
Since this remainder contains another 8 we convert our answer to **26 rem 3**.

✐ **Practice G** Try the following.

a 8) 1 1 1 **b** 8) 1 5 1 **c** 8) 1 0 0 **d** 8) 2 1 4 **e** 8) 1 1 2 1

---

a 13 r7 b 18 r7 c 12 r4 d 26 r6 e 140 r1

10 Now, in dividing by 7 which is 3 below 10 we must **treble** the last answer figure at each step.

7) 1 1
1 r 4

and

7) 1 2 3
1 5 r18 = 17 r4

✐ **Practice H** Try these:

a 7) 1 3 **b** 7) 3 1 **c** 7) 2 3 **d** 7) 4 0

e 7) 1 0 3 **f** 7) 1 1 1 **g** 7) 1 0 0

---

a 1 r6 b 4 r3 c 3 r2 d 5 r5
e 14 r5 f 15 r6 g 14 r2

15.3 DIVISION BY 99, 98 ETC.

11 Suppose we want to divide the number **121314 by 99**.

This is very similar to division by 9, but because 99 has two 9's we can get the answer **two digits at a time**.

Think of the number split into pairs: 12/13/14 where the last pair is part of the remainder.
Then put down the 12 as the first part of the answer:

```
99) 12 / 13 / 14
    12
```

Then add the 12 to the 13 and put down 25 as the next part:

```
99) 12 / 13 / 14
    12 / 25
```

Finally add the 25 to the last pair and put down 39 as the remainder:

```
99 ) 12 / 13 / 14
     12 / 25 / 39
```

So the answer is **1225 remainder 39**.

✎ **Practice I** Divide by 99:

a 121416 **b** 213141 **c** 332211 **d** 282828 **e** 363432

f 11221122 (this has 4 pairs, but the method is the same) **g** 3456 (this has 2 pairs)

---

a 1226 r42 b 2152 r93 c 3355 r66 d 2856 r84 e (3670 r102) 3671 r3
f 113344 r66 g 34 r90

Dividing by 98 is similar.

12 **121314 ÷ 98 = 1237 remainder 88**.
.
This is the same as before, but because 98 is **2** below 100 we **double** the last part of the answer before adding it to the next part of the sum.

So we begin as before by bringing 12 down into the answer:

```
98 ) 12 / 13 / 14
     12
```

Then we double 12 and add this to 13 to get 37:

```
98 ) 12 / 13 / 14
     12 / 37
```

Finally double 37 and add it to 14:

```
98 ) 12 / 13 / 14
     12 / 37 / 88   = 1237 remainder 88
```

✐ **Practice J** Divide by 98:

a 112203	**b** 102010	**c** 131313	**d** 200202	**e** 2131
a 1144 r91	**b 1040 r90**	**c 1339 r91**	**d 2042 r86**	**e 21 r 73**

In a similar way we can divide by numbers like 97 and 999.

15.4 DIVISOR BELOW A BASE NUMBER

Dividing by 9 is easy, as you have seen.
It is similarly easy to divide by numbers near other base numbers: 100, 1000 etc.

13 Suppose we want to divide **235 by 88** (which is close to 100).

We need to know how many times 88 can be taken from 235 and what the remainder is.
Since every 100 must contain an 88 there are clearly two 88's in 235.
And the remainder will be two 12's (because 88 is 12 short of 100) plus the 35 in 235.
So the answer is **2 remainder 59** (24+35=59).

A neat way of doing the division is as follows.

8	**8**	**) 2**	**3**	**5**

We separate the two figures on the right because 88 is close to 100 (which has 2 zeros).

Then since 88 is 12 below 100 we put 12 below 88, as shown below.

8	**8**	**) 2**	**3**	**5**
1	2		2	4
		2	**5**	**9**

We bring down the initial 2 into the answer.

This 2 then multiplies the flagged 12 and the 24 is placed under the 35 as shown.
We then simply add up the last 2 columns.

Note that the deficiency of 88 from 100 is given by the formula *All from 9 and the Last from 10.*

Note also that **the position of the vertical line is always determined by the number of noughts in the base number**: if the base number has 4 noughts then the vertical line goes 4 digits from the right, and so on.

This is easily understood since when we bring the initial 2 down into the answer we are expecting to find two 88's in 235. And as there is one 88 in every hundred and 12 left over, in two hundreds there will be two 88's and two 12's remainder, which must be added to the 35 to give 59 as the full remainder.

14 Divide **31313 by 7887**.

We set the sum out as before:

7 8 8 7) 3	**1**	**3**	**1**	**3**

7 8 8 7) 3	**1**	**3**	**1**	**3**
2 1 1 3	6	3	3	9
3	**7**	**6**	**5**	**2**

Applying *All From 9 and the Last From 10* to 7887 gives 2113.
Bring the 3, (the first figure of 31313) down into the answer.

We now multiply this by the flagged 2113 and put 6339 in the middle row.
Then adding up the last four columns gives the remainder of 7652.

✐ **Practice K** Divide the following (do as many mentally as you can):

a 88)1 2 1 **b** 76)2 1 1 **c** 83)1 3 2

d 98)3 3 3 **e** 887)1 2 2 3 **f** 867)1 5 1 3

g 779)2 2 2 2 **h** 765)3 0 0 1 **i** 8907)1 3 1 0 3

j 7999)1 2 3 2 1 **k** 7789)2 1 0 1 2 **l** 8888)4 4 3 4 4

a	**1/33**	**b**	**2/59**	**c**	**1/49**
d	**3/39**	**e**	**1/336**	**f**	**1/646**
g	**2/664**	**h**	**3/706**	**i**	**1/4196**
j	**1/4322**	**k**	**2/5434**	**l**	**4/8792**

TWO-FIGURE ANSWERS

Here we consider the case where the answer consists of more than one digit.

15 **1108 ÷ 79** = 13 remainder 81 = **14 remainder 2**.

We set the sum out marking off two figures on the right and leave two rows as there are to be two answer figures:

7 9) 1	**1**	**0**	**8**

7 9) 1	**1**	**0**	**8**
2 1	2	1	
		6	3
1	**3**	**8**	**1**

Bring the first 1 down into the answer.
Multiply the flagged 21 by this 1 and put the answer (2 1) as shown in the second row.

Adding in the second column we get 3 which we put down and then multiply the 21 by this 3 to get 63, which we place as shown in the third row.

Add up the last two columns, but since the remainder, 81, is greater than the divisor, 79, there is another 79 contained in 81 so there are 14 79's in 1108 with 2 remaining.

16 Find **1121123 ÷ 8989**.

8 9 8 9) 1	**1**	**2**	**1**	**1**	**2**	**3**
1 0 1 1	1	0	1	1		
		2	0	2	2	
			4	0	4	4
1	**2**	**4**	**6**	**4**	**8**	**7**

The initial 1 comes down into the answer and multiplies the flagged 1011.
This is placed as shown in the second row.

Adding in the second column we put 2 down in the answer and then multiply the 1011 by it. Put 2022 in the third row.

Adding in the third column we get 4 which we put down and also multiply by 1011. So we put 4044 in the fourth row and then add up the last four columns to get the remainder.

Once the vertical line has been drawn in you can see the number of lines of working needed: this is the number of figures to the left of this line (3 figures and therefore 3 lines of working in Example 16 above).

✐ **Practice L** Divide the following:

a 8 9)1 0 2 1 **b** 8 8) 1 1 2 2 **c** 7 9)1 0 0 1

d 8 8)2 1 1 1 **e** 9 7) 1 1 1 1 **f** 8 8 8) 1 0 0 1 1

g 8 8 7)1 1 2 4 3 **h** 8 9 9)2 1 2 1 2 **i** 9 8 8)3 0 1 2 5

j 8 8 9 9)2 0 1 0 2 0

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a	**11/42**	**b**	**12/66**	**c**	**12/53**
d	**23/87**	**e**	**11/44**	**f**	**11/243**
g	**12/599**	**h**	**23/535**	**i**	**30/485**
j	**22/5242**				

A SIMPLIFICATION

In these examples (and in the ones in the next section) the lines of working can be dispensed with by using the *Vertically and Crosswise* formula. We use the vertical and crosswise products in the flag and answer digits.

In Example 15 we have 21 flagged and the first answer figure is 1:

2 1
1 -

The first vertical product here gives 2×1=2 which is to be added in the second column of 1108 to give 3 as the second answer figure:

2 1
1 3

So now we take the cross-product 2×3 + 1×1 = 7 and add this to the 0 in 1108 to give 7 as the first remainder figure.
Finally the vertical product on the right in

2 1
1 3

gives 1×3=3 to be added to the last figure of 1108 which makes 11 and gives the full remainder of 7_11 = 81.

Similarly longer sums like Example 16 can also be dealt with in this way.

15.5 DIVISOR ABOVE A BASE NUMBER

A very similar method, but under the formula *Transpose and Apply* allows us to divide numbers which are close to but above a base number.

17 **1489 ÷ 123 = 12 remainder 13**.

Here we see that 123 is close to the base of 100 so we mark 2 figures off on the right.
In fact the method is just as before except that we write the flagged numbers as bar numbers:

1 2 3) 1	**4**	**8**	**9**
$\bar{2}\ \bar{3}$	$\bar{2}$	$\bar{3}$	
		$\bar{4}$	$\bar{6}$
1	**2**	**1**	**3**

Bring the initial 1 down into the answer.
Multiply this 1 by the flagged $\bar{2}\ \bar{3}$ and write down $\bar{2}, \bar{3}$.
Add in the second column and put down 2.
Multiply this 2 by the $\bar{2}\ \bar{3}$ and put $\bar{4}, \bar{6}$.
Then add up the last two columns.

The Sutra in use is *Transpose and* Apply, as stated above, because we are actually subtracting from the digits 4, 8 and 9.

✐ **Practice M** Divide the following:

a 1 2 3)1 3 7 7 **b** 1 3 1)1 4 8 1 **c** 1 2 1)2 5 6

d 1 3 2)1 3 6 6 **e** 1 2 1 2)1 3 5 4 5 **f** 1 6 1)1 7 8 1

g 1 0 0 3)3 2 1 9 8 7 **h** 1 1 1)7 9 9 9 9

---

a 11/24 b 11/40 c 2/14
d 10/46 e 11/213 f 11/10
g 321/24 h 720/79

Two other variations, where negative numbers come into the answer or remainder are worth noting next.

18 **10121 ÷ 113 = 89 remainder 64**.

1 1 3) 1	**0**	**1**	**2**	**1**
$\bar{1}$ $\bar{3}$		$\bar{1}$	$\bar{3}$		
			1	3	
				1	3
1	$\bar{1}$	$\bar{1}$	**6**	**4**

When we come to the second column we find we have to bring $\bar{1}$ down into the answer, multiplying this by the flagged $\bar{1}\,\bar{3}$ means we add 13 in the third row (two minuses make a plus).

The answer $1\bar{1}\,\bar{1}$ we finally arrive at is the same as 100 – 11 which is 89.

19 Find **2211 ÷ 112**.

1 1 2) 2	**2**	**1**	**1**	
$\bar{1}$ $\bar{2}$		$\bar{4}$		
		0	0	
2	**0**	$\bar{\mathbf{3}}$	**1**	**=** 20 rem $\overline{29}$ or **19 rem 83**

20 remainder –29 means that 2211 is 29 short of 20 112's.
This means there are only 19 112's in 2211, so **we add 112 to –29** to get 19 remainder 83.

Practice N Divide the following:

a 1 1 2)1 2 3 4 **b** 1 2 1)3 9 9 3 **c** 1 0 3)4 3 2

d 1 0 1 2)2 1 3 1 2 **e** 1 2 2)3 3 3 3 **f** 1 2 3)2 5 8 4

g 1 1 3)1 3 6 9 6 **h** 1 2 1 2)1 3 7 9 8 7 **i** 1 1 1)7 9 9 9 9

j 1 2 1)2 6 5 2 **k** 1 2 3 1)3 3 0 3 3

a	**11/02**	**b**	**33/00**	**c**	**4/20**
d	**21/060**	**e**	**27/39**	**f**	**21/01**
g	**121/23**	**h**	**113/1031**	**i**	**720/79**
j	**21/111**	**k**	**26/1027**		

"We go on, at last, to the long-promised Vedic process of STRAIGHT (AT SIGHT) DIVISION which is a simple and easy application of the URDHVA-TIRYAK Sutra which is capable of immediate application to all cases and which we have repeatedly been describing as the 'CROWNING GEM of all' for the very simple reason that over and above the universality of its application, it is the most supreme and superlative manifestation of the Vedic ideal of the at-sight mental-one-line method of mathematical computation."
From "Vedic Mathematics", Page 240.